

KVITFJELL, Norway

WMO: 013750

Lat: 61.465N

Lon: 10.127E

Elev: 1028

StdP: 89.57

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.1	(c) -16.3	(d) -21.7	(e) 0.6	(f) -15.2	(g) -19.5	(h) 0.8	(i) -15.7	(j) 18.1	(k) -4.7	(l) 15.6	(m) -4.1	(n) 2.3	(o) 160	(p) 0.873

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	19.9	12.6	18.0	12.0	16.1	11.1	14.0	17.8	12.9	15.9	12.0	14.8	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	10.2	14.7	11.6	9.7	13.9	10.7	9.0	12.9	42.4	18.0	39.4	15.9	36.9	14.9	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	12.8	11.1	-19.1	22.0	2.7	2.5	-21.1	23.8	-22.6	25.2	-24.1	26.7	-26.1	28.5
			-19.5	15.0	2.6	1.4	-21.3	16.0	-22.8	16.8	-24.3	17.6	-26.1	18.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-7.8	-7.2	-4.9	-0.5	4.2	8.5	11.5	9.5	5.8	0.7	-3.8	-5.8		
	DBStd	7.64	4.17	4.55	4.27	3.78	4.48	3.70	3.40	2.52	3.04	3.28	4.01	4.32		
	HDD10.0	3451	551	481	463	315	188	71	19	41	129	288	413	491		
	HDD18.3	6363	809	714	722	565	437	294	213	274	376	546	663	749		
	CDD10.0	131	0	0	0	0	9	27	66	24	4	0	0	0		
	CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.1	4.8	4.8	5.4	5.2	5.1	5.2	4.8	4.7	5.4	5.4	5.1	5.0		
(15) Precipitation	PrecAvg	694	49	32	30	36	61	74	93	96	59	63	60	41		
	PrecMax	1043	100	74	76	87	144	147	171	159	106	204	236	82		
	PrecMin	514	4	9	0	1	5	2	25	16	19	9	9	3		
	PrecStd	103	23	16	20	19	34	36	33	39	23	41	44	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.1	7.2	11.1	19.1	21.6	22.9	17.9	15.2	11.0	6.6	4.2		
		MCWB	0.9	1.6	2.6	6.0	10.3	11.7	14.5	13.0	10.8	8.8	3.7	2.5		
	2%	DB	1.1	2.1	4.7	8.6	16.1	19.2	20.8	16.3	13.2	7.9	4.5	2.6		
		MCWB	-0.5	-0.5	1.5	4.1	9.7	11.0	14.0	12.0	9.7	5.5	3.0	0.8		
	5%	DB	-0.6	0.5	3.0	6.9	13.4	16.6	19.0	14.8	11.5	6.4	2.9	1.3		
		MCWB	-1.5	-1.3	0.4	3.2	8.0	10.1	12.8	11.2	8.9	4.4	2.1	-0.3		
	10%	DB	-2.3	-1.3	0.9	5.2	11.0	14.3	17.2	13.3	10.1	5.2	1.4	-0.2		
		MCWB	-3.0	-2.5	-1.1	1.9	7.0	9.0	12.2	10.9	7.9	3.8	0.6	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.0	3.5	6.6	11.9	12.9	16.2	14.2	12.1	10.0	4.7	2.7		
		MCDB	2.5	3.6	6.1	10.0	16.5	16.7	20.9	16.3	13.7	10.6	5.6	3.7		
	2%	WB	-0.3	0.1	1.7	4.7	10.2	12.0	14.8	13.1	10.5	6.5	3.1	1.2		
		MCDB	0.8	1.5	4.2	7.5	14.5	16.5	19.4	14.7	12.1	7.3	4.1	2.3		
	5%	WB	-1.5	-1.2	0.5	3.6	8.8	11.0	13.7	12.3	9.3	5.2	2.0	-0.1		
		MCDB	-0.7	0.0	2.7	6.4	12.7	15.2	17.8	13.8	10.8	5.9	2.9	0.9		
	10%	WB	-2.9	-2.4	-0.9	2.2	7.3	9.9	12.6	11.2	8.3	4.1	0.7	-1.3		
		MCDB	-2.3	-1.3	0.7	4.8	10.3	14.0	16.1	12.8	9.8	4.9	1.4	-0.4		
(35) Mean Daily Temperature Range	MDBR		3.7	4.0	4.6	5.2	6.2	7.0	6.4	5.4	5.0	3.9	3.6	3.8		
	5% DB	MCDBR	5.1	5.5	6.1	7.0	8.9	9.2	8.2	6.9	6.7	5.6	5.0	5.6		
		MCWBR	4.5	4.4	4.4	4.3	4.7	4.7	4.1	4.0	4.3	4.2	4.2	5.1		
	5% WB	MCDBR	4.9	5.1	5.9	6.4	8.0	7.6	7.2	5.6	5.7	4.9	4.9	5.4		
		MCWBR	4.5	4.4	4.4	4.2	4.7	4.5	4.1	3.8	4.1	4.4	4.3	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.191	0.236	0.265	0.296	0.307	0.305	0.317	0.310	0.282	0.265	0.204	0.157		
	taud		2.141	2.290	2.289	2.306	2.480	2.544	2.510	2.522	2.587	2.520	2.207	1.984		
	Ebn at Noon		585	778	868	901	920	926	902	883	857	745	555	439		
	Edh at Noon		33	59	85	103	95	91	92	83	65	48	31	22		
(44) All-Sky Solar Radiation	RadAvg		0.29	0.96	2.53	4.30	4.85	5.16	4.72	3.66	2.36	1.07	0.35	0.16		
	RadStd		0.05	0.19	0.25	0.43	0.48	0.50	0.57	0.31	0.27	0.17	0.07	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page